

Volumes Using Washers

Numberbender | WORKSHEET



Name: _____ Date: _____ Score: _____

Find the volume of each region using the Washer Method. Show all work.

Calculus 1 Worksheet #50 Volumes Using Washers

Notes: Today you learn how to find volumes using **WASHERS**: $\text{Volume} = \pi \int_a^b R^2 - r^2$

Medium (Washer Method)

Find the volume of $y = x^2$, $y = 4$ revolved around x-axis.

Cross sections are **circular washers**.

[A washer is a circle with a **hole** in its center.]

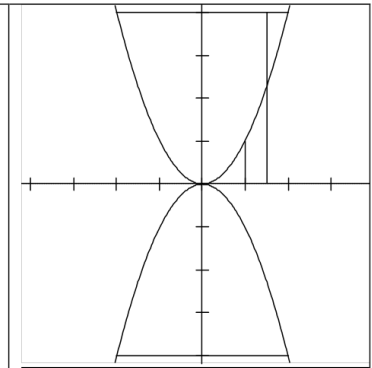
"Thickness" of **this** washer is $x_2 - x_1$, or **dx**.

$$\therefore \int_{x=-}^{x=} \pi R^2 dx - \int_{x=-}^{x=} \pi r^2 dx$$

[Volume **entire**] - [Volume "**hole**"] or

$$\pi \int_{x=-2}^{x=2} (R^2 - r^2) dx \Rightarrow \pi \int_{-2}^2 [4^2 - y^2] dx \Rightarrow \pi \int_{-2}^2 [16 - (x^2)^2] dx = \pi \int_{-2}^2 (16 - x^4) dx$$

$$\pi \left[16x - \frac{x^5}{5} \right]_{-2}^2 = \pi \left[\left(32 - \frac{32}{5} \right) - \left(-32 - \frac{-32}{5} \right) \right] = \pi \left[\frac{128}{5} - \left(-\frac{128}{5} \right) \right] = \boxed{\frac{256\pi}{5}}$$



Find the volume of the following:

1. $y = x^3$, $x = 2$, $y = 0$ (x-axis) 2. $y = 2x^2$, $y = 0$, $x = 5$ (x-axis) 3. $y = 4x^2$, $y = 16$ (y-axis)

4. $y = \frac{x}{2}$, $y = 2$ (y-axis) 5. $y = \sqrt{x}$, $x = 0$, $y = 2$ (y-axis) 6*. $y = \frac{x}{2}$, $x = 0$, $y = 2$ (x-axis)

7*. $y = \sqrt{x}$, $x = 0$, $y = 2$ (x-axis) 8*. $y = x$, $y = 3x - x^2$ (x-axis) 9*. $y = \frac{x^2}{3}$, $y = x$ (y-axis)

10. Use the Fundamental Theorem of Calculus to evaluate the integral: $\int_0^{\frac{\pi}{4}} \tan x dx =$

11. Use the Fundamental Theorem of Calculus to evaluate the integral: $\int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \sin x dx$

12. Calculate the area bounded by $y = x^3 - 4x + 4$ and $y = 3x - 2$.



Math is hard. So we made it human.
Labanan ang agwat.

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Volume by Washers Worksheet

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Volumes Using Washers

Numberbender | ANSWER KEY



Answer key — for instructor use only.

[Use your calculator to sketch (CAREFULLY) and to find intersection points] **Do the rest by hand!**

Answers:

1. $\frac{128\pi}{7}u^3$	2. $2500\pi u^3$	3. $32\pi u^3$	4. $\frac{32\pi}{3}u^3$	5. $\frac{32\pi}{5}u^3$	6. $\frac{32\pi}{3}u^3$
7. $8\pi u^3$	8. $\frac{56\pi}{15}u^3$	9. $\frac{9\pi}{2}u^3$	10. $\ln \sqrt{2}$	11. $\sqrt{2}$	12. $\frac{131}{4}u^2$

